Predicting Flight Delays Using Supervised Machine Learning

A Case Study on U.S. Domestic Airlines (2019–2023)

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***Abstract*-Flight delays are a recurring challenge for airlines, passengers, and regulators, leading to significant economic and operational consequences. This study investigates the use of supervised machine learning to predict flight delays on U.S. domestic airlines using historical schedule, route, and time-based features from 2019–2023. The dataset comprised 3.000,000 flights with 32 features, and the target variable was defined as an arrival delay of 15 minutes or more. Two models, Logistic Regression and Random Forest, were trained and evaluated under class imbalance settings, with Random Under-Sampling (RUS) applied to improve minority-class representation. Model performance was assessed using Accuracy, Precision, Recall, F1-score, and ROC-AUC. Results show that baseline Logistic Regression achieved high overall accuracy (82.4%) but completely failed to detect delayed flights (F1 = 0). Applying RUS improved detection, with Logistic Regression reaching an F1-score of 0.596 and ROC-AUC of 0.616. Random Forest with RUS achieved (F1 = 0.596, ROC-AUC = 0.637). These findings highlight the difficulty of predicting flight delays using schedule-only features and demonstrate the importance of resampling techniques in handling imbalanced classification problems.**

Keywords- flight delay prediction, machine learning, logistic regression, random forest, class imbalance, supervised learning

I. INTRODUCTION

Flight delays are a persistent challenge in the aviation industry, creating substantial inconvenience for passengers and significant economic losses for airlines. Delays increase operational costs through wasted fuel, inefficient crew scheduling, and reduced aircraft utilization, while also impacting customer satisfaction and airline competitiveness. According to the U.S. Bureau of Transportation Statistics, millions of flights are delayed each year, with knock-on effects across the entire air transportation system. Predicting whether a flight will be delayed is therefore of high importance to both business operations and public service delivery.

The central research question addressed in this work is: Can supervised machine learning models trained on schedule, route, and time-based features effectively predict flight delays in U.S. domestic airlines? This question is explored through the following hypotheses:

- Null Hypothesis (H_0): Schedule-, route-, and time-based features do not significantly predict flight delays.
- Alternative Hypothesis (H_1): Schedule-, route-, and time-based features significantly predict flight delays.

The aim of this work is to design and evaluate predictive models that classify flights as on-time or delayed based on historical data. Two widely used machine learning algorithms, Logistic Regression and Random Forest, were implemented and compared. Due to the strong class imbalance between delayed and non-delayed flights, Random Under-Sampling (RUS) was applied to improve model sensitivity to the minority (delay) class.

The key contributions of this report are as follows:

1. Development of a full pipeline for flight delay prediction using schedule-based airline data.
2. Comparative evaluation of Logistic Regression and Random Forest models.
3. Empirical demonstration that class balancing techniques improve the ability to detect delayed flights.
4. Interpretation of predictive features to provide operational insights for airline delay management.

The proposed solution leverages supervised machine learning to identify patterns in historical airline operations and evaluate their predictive power. By benchmarking Logistic Regression against Random Forest, this work contributes an interpretable and replicable framework for understanding delay risks and guiding operational decision-making in the airline industry.

II. LITERATURE REVIEW

Recent studies approach flight-delay prediction with classical ML, deep learning, and hybrid pipelines, typically built on U.S. Bureau of Transportation Statistics (BTS) data and schedule/operational features. Gholami and Khashe developed ML/DL models for U.S. domestic flights and emphasized the operational link between departure and arrival delays, also proposing a conversational (voice) interface for real-world use [1]. Reviews from 2019–2024/25 consistently report that Random Forests are strong for binary delay classification, LSTM-type models help when exploiting temporal sequences, and that class imbalance and real-time constraints remain persistent challenges; they also point to hybrid and spatiotemporal formulations to capture delay propagation across airports [2].

Comparative single-dataset papers frequently benchmark RF, KNN, and SVM against linear baselines and find tree ensembles competitive on heterogeneous airline features,

though performance depends on label definition and feature scope [3]. Weather-focused systems show that meteorological variables (e.g., temperature, humidity, visibility, rainfall) are predictive but do not fully explain delays when carrier and NAS (airspace) factors dominate in many samples—motivating feature sets that include schedule/route/operations alongside weather [4]. A recent hybrid ML–DL case study on U.S. airlines evaluates models with Accuracy, Precision/Recall/F1, ROC–AUC, and extensive EDA, underscoring the need to look beyond accuracy in imbalanced settings [5].

Guided by the above, this work intentionally tests how far schedule, route, and time-based predictors (without external weather feeds) can go. In line with prior findings on imbalance, a baseline Logistic Regression achieved high accuracy but has trouble to detecting delayed flights, while resampling improved minority-class detection; overall discrimination remained modest without richer features, which is consistent with survey guidance to address imbalance and broaden features or adopt propagation-aware/hybrid models [2]– [5].

III. METHOD DESCRIPTION

A. The Dataset

This study uses U.S. domestic flight records from 2019–2023, comprising schedule, route, and calendar attributes [6]. The original data set consists of 32 columns and 3,000,000 rows (Table I).

No.	Variable	Description	Data Type
1	FL_DATE	Flight date	object
2	AIRLINE	Airline name	object
3 - 4	Codes related to AIRLINE	Airline DOT and IATA code	object
5	DOT_CODE	Airline DOT numeric code	int64
6	FL_NUMBER	Flight number	int64
7 - 8	ORIGIN, ORIGIN_CITY	Origin airport code and City	object
9 - 10	DEST, DEST_CITY	Destination airport code and City	object
11	CRS_DEP_TIME	Scheduled departure time (hhmm)	int64
12	DEP_TIME	Actual departure time (hhmm)	float64
13	DEP_DELAY	Departure delay in minutes	float64
14	TAXI_OUT	Time spent taxiing out (minutes)	float64
15	WHEELS_OFF	Time aircraft wheels left the ground (hhmm)	float64
16	WHEELS_ON	Time aircraft wheels touched down (hhmm)	float64
17	TAXI_IN	Time spent taxiing in (minutes)	float64
18	CRS_ARR_TIME	Scheduled arrival time (hhmm)	int64
19	ARR_TIME	Actual arrival time (hhmm)	float64
20	ARR_DELAY	Arrival delay in minutes	float64
21 - 23	Cancel/ Diverted data	Flight Cancel/ Divert or not binary, cancellation code	float64, Object
24	CRS_ELAPSED_TIME	Scheduled elapsed time (minutes)	float64
25	ELAPSED_TIME	Actual elapsed time (minutes)	float64
26	AIR_TIME	Time spent in the air (minutes)	float64
27	DISTANCE	Distance between origin and destination (miles)	float64
28 - 32	DELAY_DUE_Resons	Delay minutes due to Reasons	float64

Table I: Description and data types of variables

After preprocessing, the working table contains 2,913,802 rows and 32 columns. The binary target, delay status, follows the industry "ArrDel15" convention: a flight is labelled delayed when the arrival delay is at least 15 minutes [1]. In our pipeline, this was implemented as ARR_DELAY > 15; with integer-minute delays, this is effectively equivalent to ≥15.

In preprocessing, the flight date was converted to a proper datetime type, from which day-of-week and month were derived. To capture seasonality without discontinuities at cycle boundaries, both were transformed into cyclical encodings (e.g., month_sin, month_cos, and day_sin). Scheduled and actual clock fields provided in HHMM format were normalized to fractional hours; we retained scheduled departure hour, scheduled arrival hour and scheduled elapsed time for modelling.

For missing data and exclusions, we removed rows for cancelled and diverted flights. Columns with very high missing data like cancellation code, delay due carrier, delay weather, delay due security and delay due to late aircraft, were excluded from the modelling matrix (these were only used for descriptive plots of delay causes). Any remaining rows with missing values were dropped, especially because we couldn't impute them due to high missing values. The final encoded dataset contains no missing values.

Categorical fields airline code, origin and destination factorized. Numeric distributions were inspected with boxplots; no explicit winsorization/clipping was applied

because schedule/route features are naturally bounded, and the tree-based model is robust to moderate skew. A correlation heatmap on selected numeric variables (Figure 1) showed strong associations among schedule variables (e.g., schedules departure time, scheduled arrival time, actual elapsed time), and a high correlation between distance and elapsed time, informing feature selection and interpretation.

Schedule variables are strongly correlated (e.g., planned departure/arrival hour and scheduled elapsed time), and distance aligns closely with scheduled elapsed time. These relationships informed feature selection and interpretability considerations for the models.

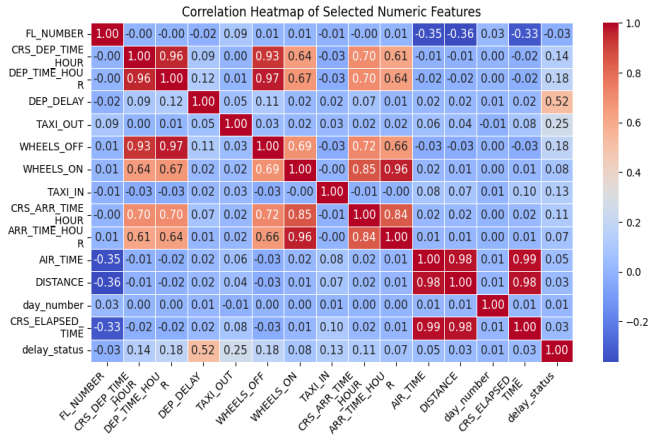

Figure 1: Correlation heatmap of selected numeric features

After considering the possible multicollinearity and objective of the study a set of schedule/route related features has been selected for the model: flight number, season of year planned departure and arrival hours, planned flight duration, route distance, airline IATA code, Origin Airport code, destination airport code and day-of-week. This choice aligns with our goal to assess how far schedule/route/time signals alone can predict delays, without external weather or network-propagation inputs.

However, the data set is highly imbalanced, and finally, we balanced the data set with the Random Under-Sampling technique.

B. *Data Mining Method*

We formulate flight–delay prediction as a binary classification task (delayed vs. on time, using the ≥15-minute arrival threshold). Models are trained on the compact, operational feature set described in Section III-A (flight number; seasonal encodings of month/day; scheduled departure/arrival hours; scheduled elapsed time; and route distance). To avoid target leakage, no realized post-departure variables are used.

Two supervised learners are evaluated. Logistic Regression serves as an interpretable baseline, estimating the probability of delay with an L2-regularized linear decision function. Random Forest serves as a non-linear comparator that aggregates multiple decision trees to capture interactions among schedule and route features. For both models, we prioritize a fair, like-for-like comparison on the same predictors; hyperparameters are kept standard (e.g., LR with L2 regularization and a conventional solver; RF with a typical number of trees and default depth constraints), since our focus is on the effect of class balancing rather than exhaustive tuning.

Because delayed flights are the minority class, we assess performance under two training regimes: (i) Logistic Regression on the original imbalanced training split, and (ii) Logistic Regression with Random Under-Sampling (RUS) and Random Forest with RUS. Under-sampling is applied only to the training partition to achieve approximately 1:1 class balance; the test set remains untouched to provide unbiased estimates.

Training follows a hold-out protocol with a fixed random seed. After fitting, predicted probabilities are thresholded at 0.5 to produce class labels. Performance is summarized using Accuracy, Precision, Recall, F1-score, and ROC-AUC (curves omitted; values reported numerically). Given the class imbalance, Recall and F1 for the delay class are emphasized. As reported in Table II, the imbalanced Logistic Regression attains high apparent accuracy but zero F1 (predicting only the majority class), whereas applying RUS enables both models to detect delayed flights, with Logistic Regression + RUS yielding the strongest F1 (0.596) and both RUS models exhibiting similar ROC-AUC (~0.61).

IV. Results and Analysis

We evaluate three configurations: Logistic Regression (LR) on the original imbalanced training data, Logistic Regression with RUS and Random Forest with RUS, to test whether schedule/route/time features can predict delays (H_1).

Table II summarizes test-set performance. The imbalanced LR achieves high apparent accuracy (82.4%) but Precision = Recall = F1 = 0, indicating it predicts only the majority "on-time" class and fails to detect any delays. After balancing, Logistic Regression with RUS reaches Precision of 0.587, Recall of 0.606, F1 of 0.596, Accuracy of 58.9%, and ROC-AUC of 0.616. Random Forest with RUS shows almost a similar performance with an F1 of 0.596, Accuracy of 60.1% & ROC-AUC of 0.637). Thus, both balanced models deliver similar F1, with Random Forest with RUS showing higher AUC and accuracy. In short, accuracy alone is misleading under imbalance, and resampling is necessary to recover minority-class detection.

Model	**Comparison of Logistic Regression and Random Forest Models**				
	Accuracy	***Precision***	***Recall***	***F1-Score***	***ROC-AUC***
Logistic Regression	82.4%	0.000	0.000	0.000	0.614
Logistic Regression with RUS	58.9%	0.587	0.606	0.596	0.616
Random Forest with RUS	60.1%	0.603	0.590	0.596	0.637

Table II: Comparison of Logistic Regression and Random Forest Models

For the 3 confusion matrices, the imbalanced LR matrix (Figure 2) contains only true negatives and false negatives, which explains the zero F1 despite 82.4% accuracy. With RUS, both Logistic Regression and Random Forest (Figures 3–4) produce non-zero true positives alongside more false positives, the expected precision–recall trade-off when improving minority-class recall.

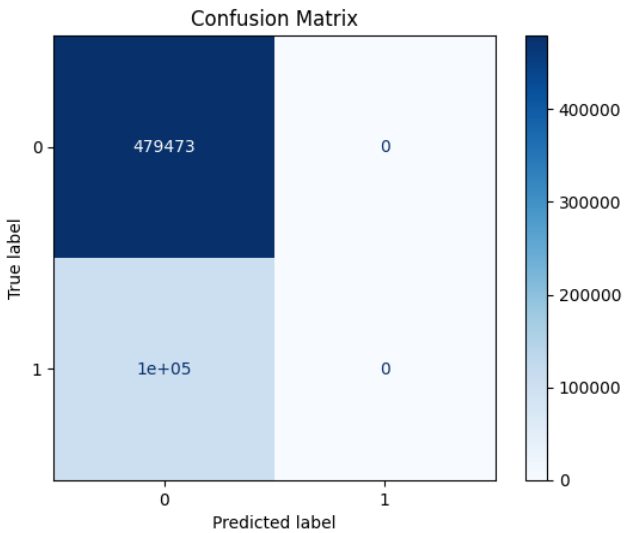

Figure 2: Imbalanced Logistic Regression Confusion Matrix

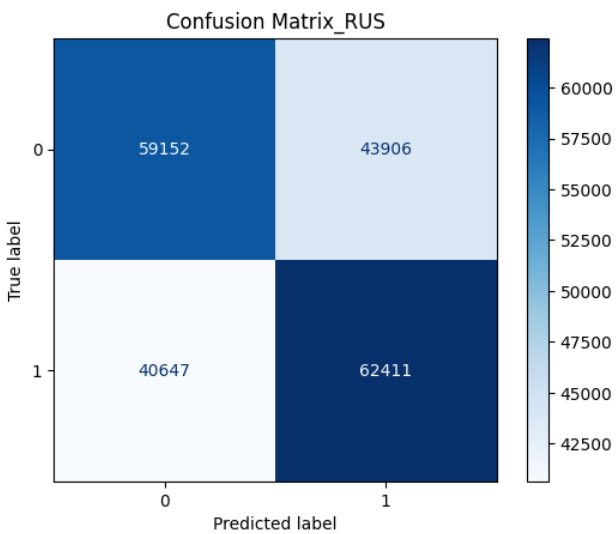

Figure 3: Logistic Regression with RUS Confusion Matrix

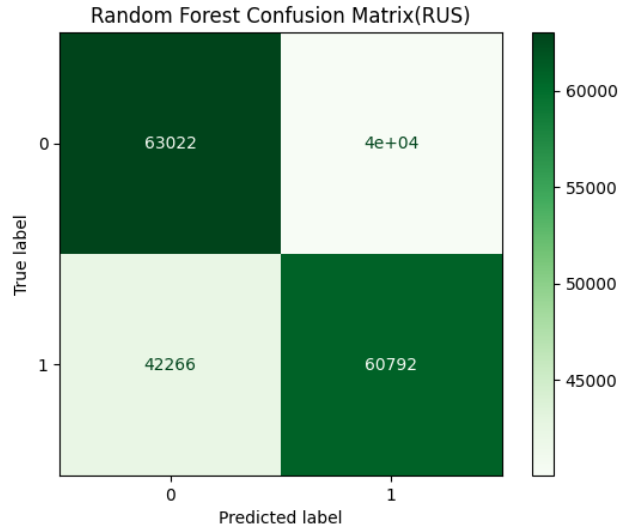

Figure 4: Random Forest with RUS Confusion Matrix

In terms of feature importance, random forest highlights schedule and route descriptors as the dominant signals: planned arrival/departure hours, flight number, planned elapsed time, and distance contribute most. This is consistent with schedule-driven risk patterns reported in the literature. We note that flight number (FL_NUMBER) can capture persistent "template" effects for recurring flights, which may boost fit but reduce portability to unseen flight numbers; this trade-off is considered in Section VI.

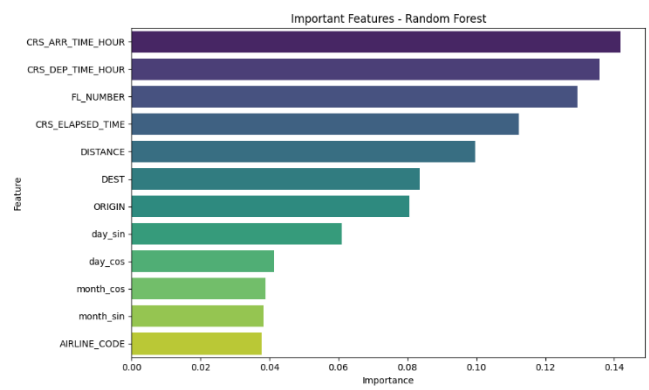

Figure 5: Feature importance plot in Random Forest

The results of 5-fold cross-validation (Table III) demonstrate consistency in both models, with Accuracy scores ranging from 58.6% to 58.8% and an average of 58.7. The mean accuracy of the 5-Fold cross-validation method of RF model is 59.9%, with a range of 59.9% to 60.1%. This indicates stable performance across different data splits.

Model	Accuracy Score					Mean
	1st Fold	2nd Fold	3rd Fold	4th Fold	5th Fold	
Logistic Regression with RUS	58.7 %	58.8 %	58.8%	58.6%	58.7%	58.7%
Random Forest with RUS	60.0 %	59.8 %	60.1%	59.9%	59.9%	59.9%

Table III: Accuracy Scores of 5-Fold cross validation

Delays display a clear seasonal pattern: counts rise sharply in June-August and fall in September-November as in Figure 6. This seasonality supports the inclusion of month as a cyclical feature (sine/cosine) and indicates that temporal effects are an important driver of delay risk.

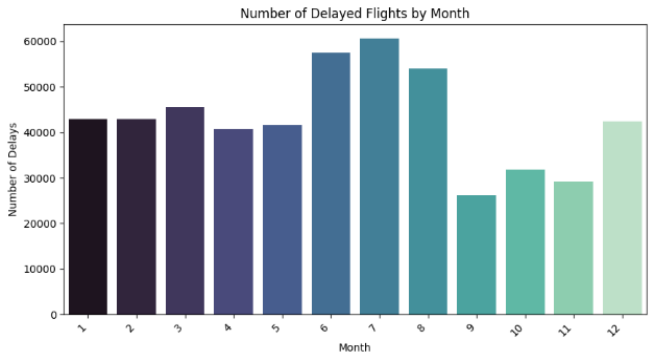

Figure 6: Flight Delays by Month

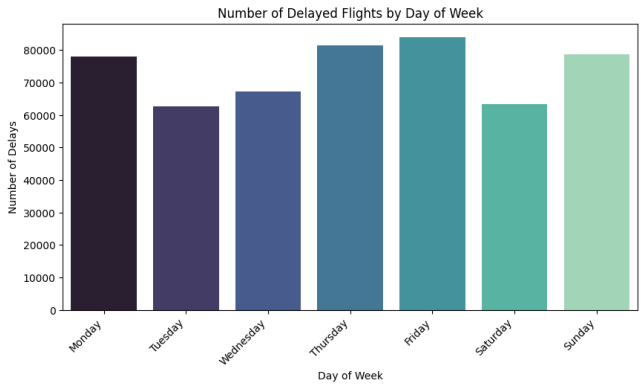

Figure 7: Flight delays by Day of Week

Delays by airlines are higher mid-week, with the largest volumes on Thursday and Friday, and lower on Saturday as in Figure 7 above. These weekday effects motivate day-of-week encodings and suggest that schedule density and operational load contribute to observed delays.

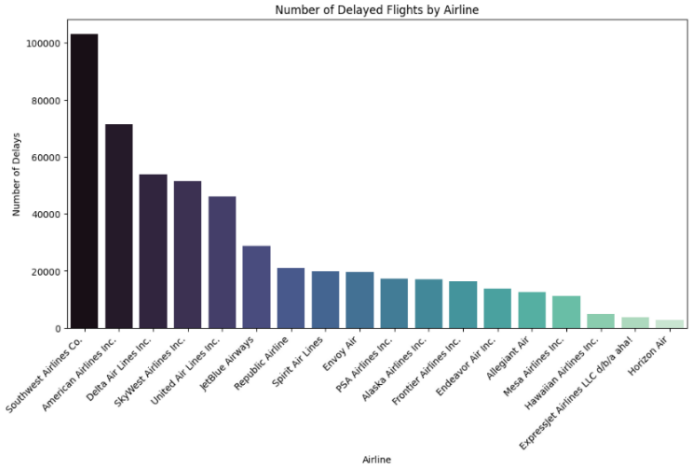

Figure 8: Flight delays by Airline

Southwest Airlines Co., American Airlines Inc., and Delta Air Lines Inc. had the highest number of delays, while Horizon Air had the least according to Figure 8 above.

Lastly, Carrier and National Airspace System (NAS) categories account for most delays (Figure 9), whereas weather and security appear less frequently in this sample. This composition justifies focusing on schedule/route/time features and explains why schedule-only models capture non-trivial signal even without external weather inputs.

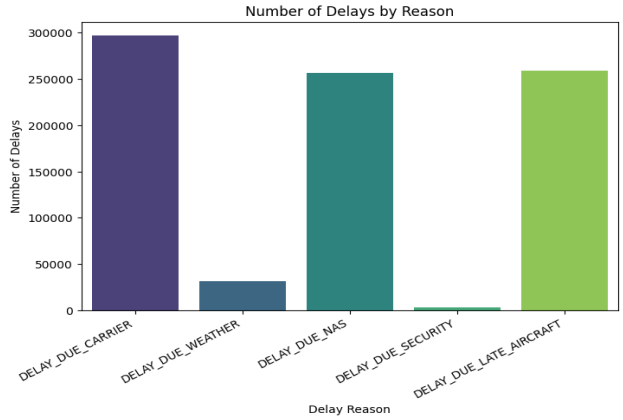

Figure 9: Delays Causes Distribution

V. CONCLUSION

This study assessed whether schedule/route/time features can predict U.S. domestic flight delays using Logistic Regression (LR) and Random Forest (RF). On the imbalanced data, LR achieved high apparent accuracy (82.4%) but zero F1 for the delayed class, illustrating the pitfalls of accuracy under class imbalance. Applying Random Under-Sampling (RUS) restored minority-class detection: Logistic Regression with RUS delivered the accuracy 58.9%, F1 (0.596) and ROC-AUC 0.616, while Random Forest with RUS achieved F1 0.596 with ROC-AUC 0.637 at the accuracy (60.1%). Feature analysis indicated that planned arrival/departure hours, scheduled elapsed time, distance, and flight number were the

most influential predictors. Descriptive patterns, seasonality and weekday effects, were consistent with these operational signals. Overall, the results partly support the hypothesis: schedule/route/time variables contain predictive signal, but on their own yield modest discrimination.

For future work, performance could be improved by (i) imbalance-aware learning beyond RUS (e.g., SMOTE, class-weighted losses, threshold tuning), (ii) incorporating weather and network/airport congestion features to capture exogenous drivers, and (iii) stronger validation (e.g., grouping by route/flight number, temporal splits) and calibration tailored to airline cost trade-offs.

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